

Assignment 7- NumPy, Pandas & Matplotlib Basics

Deadline: **26.01.26**

- All completed task files must be pushed to the designated GitHub repository.
- **Assessment:** Successful submission requires **GitHub push and defense**

Installation & Setup Instructions(5 points)

1. Install Jupyter Notebook

Install Jupyter Notebook by following the official guide:

<https://jupyter.org/install>

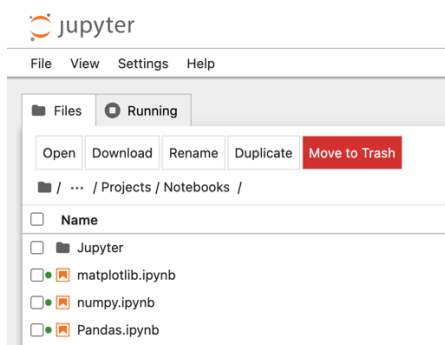
Verify installation:

jupyter notebook

2. Project Structure

1. Create a main project folder
2. Inside it, create a folder named **Notebooks**
3. Create separate .ipynb files for each task inside the **Notebooks** folder

Example structure:



Assignment-7/

└─ Notebooks/

 └─ numpy_tasks.ipynb

└─ pandas_tasks.ipynb
└─ matplotlib_tasks.ipynb

3. Install NumPy (inside Jupyter Notebook)

Run the following command in a Jupyter cell:

```
!pip install numpy
```

Verify installation:

```
import numpy as np
```

4. Install Pandas

In a Jupyter cell, run:

```
!pip install pandas
```

Verify:

```
import pandas as pd
```

5. Install Matplotlib

In a Jupyter cell, run:

```
!pip install matplotlib
```

Verify:

```
import matplotlib.pyplot as plt
```

Tasks (10 points):

Part 1: NumPy (Numerical Computing)

Task 1: Create Arrays

- Create a 1D NumPy array with numbers from 1 to 10
- Create a 2D NumPy array (3×3) with random integers

Task 2: Array Operations

- Find the sum, mean, and maximum value of the array
- Multiply all elements by 2
- Replace all values greater than 5 with 0

Part 2: Pandas (Data Analysis)

Task 3: Create a DataFrame

- Create a DataFrame with columns: Name, Age, Score
- Add at least 5 students with sample data

Task 4: Data Exploration

- Display the first 3 rows
- Show summary statistics using describe()
- Find the average score

Task 5: Data Filtering

- Select students older than 20
- Select students with a score above 80

Part 3: Matplotlib (Basic Visualization)

Task 6: Line Plot

- Create a line plot showing student scores
- Label the x-axis and y-axis
- Add a title to the plot

Task 7: Bar Chart

- Create a bar chart of student names vs scores

Example: Line chart in Jupyter Notebook using Matplotlib.

